IT Hasła - Jacek

—

Principles

SOLID

ACID

ACID 2.0

KISS

OBJECT ORIENTED PROGRAMMING PRINCIPLES

Java

Basics

Java core

Object. Wait. Notify.
Equals and hashcode, type of hashcode, duplicates of hashcodes
How to create String not in String pool, and how to move string to pool from eden
java.util.Observable
Java NIO
Reader vs InputStream
Generics
Why String is static in java?

What it is recommended to use buffered IO Streams as opposed to *Scanner* and *PrintWriter*

Possible memory leaks -> caches, event listeners & handlers, class that manages its own memory
Equals contract -> (reflexive, transitive, symmetric, consistent, equals to null returns false), equal objects must have equal hash codes
How to make class immutable
What is defensive copy?
Image that you want to dump a big database table to a file, and how to make this job faster? -> Writing it sequentially is likely to be the fastest. Writing to a normal file by definition is a serialized operation. You gain no performance by trying to write to it from multiple threads, I/O is a finite bounded resource at orders of magnitude less bandwidth than even the slowest or most overloaded CPU.

Java 8

Features

http://www.journaldev.com/2389/java-8-features-with-examples
https://www.javacodegeeks.com/2014/05/java-8-features-tutorial.html
http://www.oracle.com/technetwork/java/javase/8-whats-new-2157071.html
LocalDate
LocalDateTime
Lambdas and Functional Interfaces
Interface's Default and Static Methods
Method References
Repeating annotations
Better Type Inference
Extended Annotations Support
Optional
Streams
Nashorn JavaScript engine
Base64
Parallel Arrays
Concurrency
Class dependency analyzer: jdeps
Metaspace
Java8 stream jak skonwertować Listę do mapy
Java8 stream jak skonwertowac Listę do Stringa rozdzielonego przecinkami
Jak działa reduce
Jakie znam features'y z javy 8
Kiedy optional-a
Consumer, Supplier, Function, Predicate
Rożnica między map a flat map

Java 9

Features

https://www.baeldung.com/new-java-9

https://www.pluralsight.com/blog/software-development/java-9-new-features

The Java Platform module system

Linking

JShell: the interactive Java REPL

Improved Javadoc

Collection factory methods

Stream API improvements

Private interface methods

HTTP/2

Multi-release JARs

Process API

Variable Handles

Publish-Subscribe Framework

Unified JVM Logging

Java 10

Features

Java 11

Features

https://winterbe.com/posts/2018/09/24/java-11-tutorial/

Java 12

Features

Security

PKI

PKCS 7 - pozwala podpisywac i szyfrowac dane kluczem publicznym, BEZ klucza prywatnego
PKCS12, (.p12, .pfx) - na ogól trzymamy klucz pryw z towarzysz. Cert. klucza publicznego
PKCS#10 (.csr) - format for certificate requests
PEM - moze zawierac w sumie wszystko, lacznie z requestami
JKS
How algorithms works
Public key cryptography
Symmetric Cryptography
SSL nawiązywanie połączenia jak działa
What is keystore
Cross-site Request Forgery (**CSRF**) (naklonienie kogos zeby klinął na linka),
Cross-site Scripting (XSS) - treść przekazaną od użytkownika jako kod HTML serwisu

Attacks
Man in the middle
DES
AES (Advanced encryption standard, następca DES, symetryczny)
X.509 - certyfikat kucza publicznego, sklada sie z konteneru, info o algorytmie oraz pospisu

Threads

Thread. Run. Start.
Synchronized keyword
Cyclic Barrier
CountDownLatch
Thread starving
ThreadLocal
Stream.parallel
Callable vs Runnable
Executors
Future
ForkJoinPool.commonPool() (global, general purpose pool introduces in JDK 8).
CompletableFuture thenApply,exceptionally
Concurrency in Java 8
Rx how vs cold stream

Java Memory

Structure and flow
Heap vs Stack
Young vs Old vs Permanent Generation (or Metaspace)
Xms Xmx Xss
PermGen MetaSpace
Strong vs weak vs soft reference
-XX:+HeapDumpOnOutOfMemoryErrors
String.intern() and strings pool mngmt

Databases

Idempotency - function is idempotent if the system state remains the same after one or several calls
Serializable isolation, Snapshot isolation

Select for update, Table and rows blocking Inny link
Optimistic vs pessimistic locking -> Optimistic vs Pessimistic locking
How indices are stored
Understanding B-tree
Sql join types
Default join ->
Isolation levels - differences
Drop, Truncate - differences, TRUNCATE DELETE comparison Różnica delete truncate
DDL and DML, DCT, TCL
DML, DDL, DCL, TCL
Transactions
When single conversation with several transactions will be atomic? ->
High Water Marks
Kiedy indeks będzie używany %bb

ACID

ACID 2.0

JPA

Hibernate main classes ->
Is Hibernate Session threadsafe? ->
Hibernate Automatic Versioning
Hibernate Detached Objects
Hibernate Extended Session
Hibernate unit of work patterns ->

Hibernate batch
TransactionAttribute types
Hibernate get vs load (vs find & getReference in JPA)
session-per-request-with-detached-objects
Session-per-conversation
generationType auto vs sequence vs identity
Hibernate - cache levels ->
When does cache work
@Mapped crown????
Hibernate - how to force a version check on data you are not updating ->

Hibernate - how to convert field to some value object? ->

What is LOB? -> Database large object
Hibernate - dziedziczenie - Inheritance types -> MappedSuperclass, Single table (default), Joined table (table per subclass),Table per class

Hibernate

SQL

NoSQL

Frameworks

Spring

2 core features - Dependency Injection, Aspects
AOP
Type of Beans, Singleton, Prototype (spring - signleton, cdi - default)
Spring @Qualifier, @Profile
Proxy modes
Provider
@Value placeholder, Environment
scoped proxy mode- jakie są, jaki default (no, default, interfaces, target class)
@Component is a generic stereotype for any Spring-managed component
`@Repository` is used for translating your unchecked SQL exception to Spring Excpetion and the only exception you should deal is `DataAccessException`
Scheduling
Unit of Work
Spring default scope
Spring scopes & comparision with CDI scopes
Spring DeferredResult
`@RequestMapping`
@Suspended, AsyncResponse
Spring - jpa
Szukanie po liscie, jak powinna sie nazywac metoda
Kiedy async lub transactional zostanie zignorowane
Jakie moduły springa używasz
Typy beanów
Czym sie rozni @Component od @Repository
Co to jest SpEL
`${expr}`, #{expr} różnice --> Immediate Evaluation, Deferred Evaluation

Angular

Tests

Basics

JUnit

Mock

A mock is a dummy class replacing a real one, returning something like null or 0 for each method call. You use a mock if you need a dummy instance of a complex class which would otherwise use external resources like network connections, files or databases or maybe use dozens of other objects. The advantage of mocks is that you can isolate the class under test from the rest of the system.

Spy

A spy is kind of a hybrid between real object and stub, i.e. it is basically the real object with some (not all) methods shadowed by stub methods. Non-stubbed methods are just routed through to the original object. This way you can have original behaviour for "cheap" or trivial methods and fake behaviour for "expensive" or complex methods.

Stub

A stub is also a dummy class providing some more specific, prepared or pre-recorded, replayed results to certain requests under test. You could say a stub is a fancy mock. In Spock you will often read about stub methods.

Tools

IntellJ

Docker

Jenkins

GitLab

Jira

Spock

Swagger

Lombok

Maven

Gradle

Archiva

UML

Diagrams

Relations

Design patterns

https://magazynt3.pl/kurs-uml-czesc-1-wstep-i-diagramy-klas/
Związki, interfejsy, diagramy klas, diagramy użycia, diagramy czynności
typy modeli w UML -> strukturalne /(Class diagram

- Package diagram
- Object diagram
- Component diagram
- Composite structure diagram
- Deployment diagram

, behavioralne, architektoniczne??
Czym się różni kompozycja od agregacji? -> w kompozycji obiekt nie może istnieć bez drugiego, np. departament bez firmy

Creational patterns -> abstract factory, builder, factory method, singleton, prototype, object pool
Structural->adapter, bridge, composite (tree), decorator, facade, flyweight, private class data, proxy
Behavioral->chain of responsib, command, interpreter, iterator, mediator, memento, null object, observer, state, strategy, template method, visibility
list 3 of all kinds and describe
Adapter vs Bridge ->
J2EE patterns - http://www.oracle.com/technetwork/java/catalog-137601.html

http://www.corej2eepatterns.com/presos/CoreJ2EEPatterns2.pdf
Bounded Context
Saga pattern
Cloud native
Twelve factor app
SOLID
KISS, DRY
Separation of concerns
Leaky abstractions
Code smell
Memoization
Composition over inheritance
Gorilla-banana problem
Communicational - functional cohesion
Functional approach - applying operator to object without modifying it (returning copy)
Defensive copy
Easist way to achieve thread safety -> immutable objects

Javascript

Basics

What is npm
ES6
JSX
strict and type-converting equality comparison
remove a particular element from an array in JavaScript?
Co robi array.splice?
Lodash `_.pull` `_.remove` `_.without`
`Lodash distinct - jaka metoda`
difference between tilde(~) and caret(^) in package.json?
PWA, Service Workers,
Flash of unstyled content - FOUC
CORS - co to?
Html vs xhtml
ARIA
cookie sessionstorage and localstorage - różnice
sessionstorage - czy jest czyszczone po zamknięciu?
Co dłużej trzyma dane - localstorage czy sessionstorage?

Microservices

Basics

Co można użyć do globalnej konfiguracji

Jakie mogą być zagrożenia związane z Microserwisami (circuit breaker, security, debugging, zarzadzanie)

Sidecar - Process that encapsulates required technologies (e.g. logging, monitoring, service discovery, etc.) in the microservices environment and makes them available to microservices that are not able to use those technologies natively

Immutable Server

Zookeeper - Provides configs in a hierarchical key-value store that is replicated and synchronized across large distributed systems

Zuull

Eureka

Etcd

Spring Cloud Config

Service Discovery (DNS, Eureka, Consul)

Load Balancing

Security

Service virtualization - co to, jakie znasz? Np. WireMock

Consumer Driven Contract testing - CDC

Development Methodologies

Architecture

Emergent design

Command Query Responsibility Segregation

Domain driven design (Bounded Context, Continuous Integration (inside bounded ctx) , Continuous refactoring, Ubiquitous Language, Entity, Value Object, Aggregate, Service, Repository, Factory - treat group of object logically as one, Service, Repository), Constraint, Process, Specification (which can be compound), Context Map (a document showing bounded

contexts and relations between them), Shared Kernel, Consumer-Supplier, Separate ways, Open Host Services, Anicorruption Layers
Layered Architecture: ui, application - (no business logic or state here), domain, infrastructure (each should depend only on layers below)
Event sourcing - Magazyn event store, Zadaniem ES jest bowiem odtwarzanie aktualnego stanu aplikacji (patrz obiektów domenowych) na podstawie zdarzeń składowanych w magazynie danych zwanym Event Store. Z początku może wydawać się to rozwiązaniem bezsensownym jednak takim nie jest, ponieważ na podobnej zasadzie działają chociażby systemy bankowe.
C4
Pluses of event sourcing - assimetric scalability towards read/write, automatic audit, teamwork share
Stream processing
Reactive programming
Functional programming
OLAP
OLTP
Event-Driven Architecture
Eventual consistency vs strong consistency
Optimistic (lazy) replication
Logical clock/Vector clock
BASE - Basic Availability, Soft State and Eventual Consistency
YAGNI, KISS
Service-oriented Architecture
Horizontal vs Vertical Scaling
Imperative vs Functiona vd declarativel
OrientDB -> Live Query
AOP makes it easy to factor out technical concerns (such as security, transaction management, logging) fro
Fear of over-engineering

Web

Rest

Web services

HTTP codes
Polling vs Long Polling vs SSE vs Websocket
Co to jest etag w headerze http?
What are indempotent/or safe methods - using GET or HEAD on a resource URL, should NEVER change the resource
Dobre praktyki
Kiedy PATCH
Jak zrobić performance testy

JSON Web token - jak zablokować
Ataki DDOS - jak zablokować
Testy performance
Kiedy OPTIONS uzywamy, jak mozna wykluczyc

DDD

Knowledge sources

udemy

websites

blogs